

#8.6

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | x_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | x_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F^2 = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.
- b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | x_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i | x_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE_i + e_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

a. $N=1000$, $SSE_M = 97161.9174$, $\hat{\sigma}_F^2 = 12.024$, $\hat{\sigma}_M^2 = \frac{97161.9174}{1000-4} = 9.2138$

① $\begin{cases} H_0: \sigma_M^2 = \sigma_F^2 \\ H_a: \sigma_M^2 \neq \sigma_F^2 \end{cases}$ ② $\alpha = 0.05$ ③ $F = \frac{\hat{\sigma}_F^2}{\hat{\sigma}_M^2} \stackrel{H_0}{\sim} F(423-1, 577-1)$

④ $RR = \{ F \geq f_{0.05}(422, 576) = 1.0007 \}$ ⑤ $F_0 = \frac{12.024}{9.2138} \approx 1.305$

⑥ $\because F_0 < RR$, 故 reject H_0 #

#8.16(a)

95% CI for KIDS (OLS)

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	Variable	Lower_95	Upper_95

1	kids	-135.330	-28.323

Holding income and age constant, the 95% confidence interval for the effect of one additional child on miles traveled is **-135.33 ~ -28.323** miles.

#8.16(b)



The "Residuals vs INCOME" plot exhibits a funnel shape (the spread of residuals increases as income increases). This pattern strongly suggests the presence of **heteroskedasticity**.

#8.16(c)

Goldfeld-Quandt Test Results		
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Test	GQ_Statistic	p_value

1 Goldfeld-Quandt	3.104	0.00000

H0: Homoskedasticity (Error variances are equal).

H1: Heteroskedasticity (Error variance increases with income).

Conclusion: The GQ test yields a p-value of **0.00000**. Because the p-value is less than 0.05, we reject the null hypothesis and conclude that heteroskedasticity is present at the 5% significance level.

#8.16(d)

Comparison of OLS Confidence Intervals		
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Model	Lower_95	Upper_95

OLS Default CI	-135.330	-28.323
kids OLS Robust CI	-139.323	-24.330

#8.16(e)

GLS vs OLS Confidence Intervals		
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Model	Lower_95	Upper_95

1 OLS Robust (from d)	-139.323	-24.330
2 GLS Conventional	-119.894	-33.718
3 GLS Robust	-121.413	-32.199
